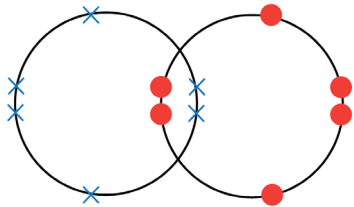


Higher Tier only questions

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		sodium is above hydrogen in reactivity series / sodium is more reactive than hydrogen / hydrogen is below sodium in reactivity series / hydrogen is less reactive than sodium	1			1		
		(ii)		$2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ accept 2e		1		1		
		(iii)		award (1) for any of following - sodium hydroxide is formed / present - hydroxide is formed OH^- ions are formed sodium hydroxide is a (strong) alkali (1)		2		2		2
	(b)	(i)		Cu^{2+} / copper ions gain (two) electrons accept $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	1			1		
		(ii)		use copper electrodes / use copper anode (1) award (1) for explanation Cu^{2+} ions coming out of solution are replaced - number of Cu^{2+} ions present (in solution) stays the same - concentration of Cu^{2+} ions (in solution) stays the same			2	2		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		 award (1) for two shared pairs award (2) for two full octets		2		2		
				Question 4 total	2	5	2	9	0	3